

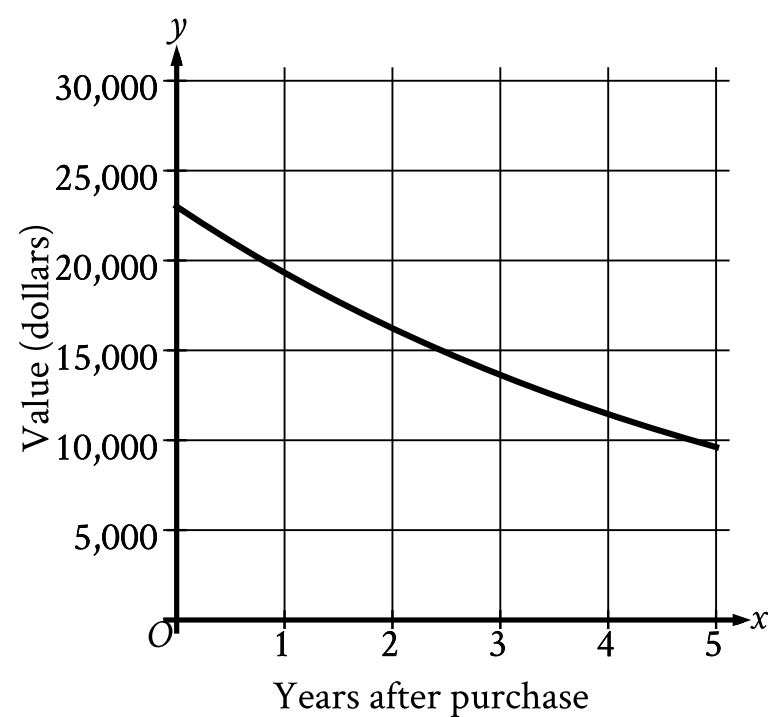
Question ID ca4ee54e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ca4ee54e

SAT5 M 2.1

The graph shows the predicted value y , in dollars, of a certain sport utility vehicle x years after it is first purchased.



Which of the following is closest to the predicted value of the sport utility vehicle **3** years after it is first purchased?

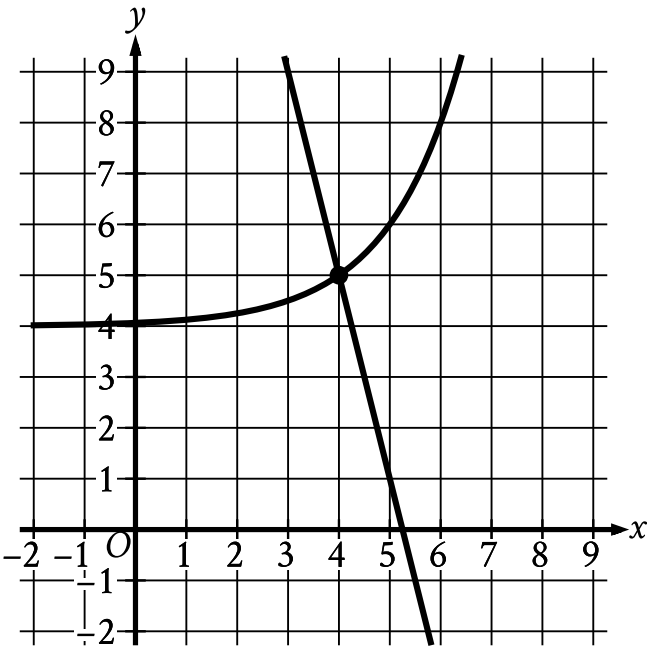
- A. \$9,619
- B. \$13,632
- C. \$19,320
- D. \$23,000

Question ID 3f8d5876

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 3f8d5876

SAT5 M 2.2



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

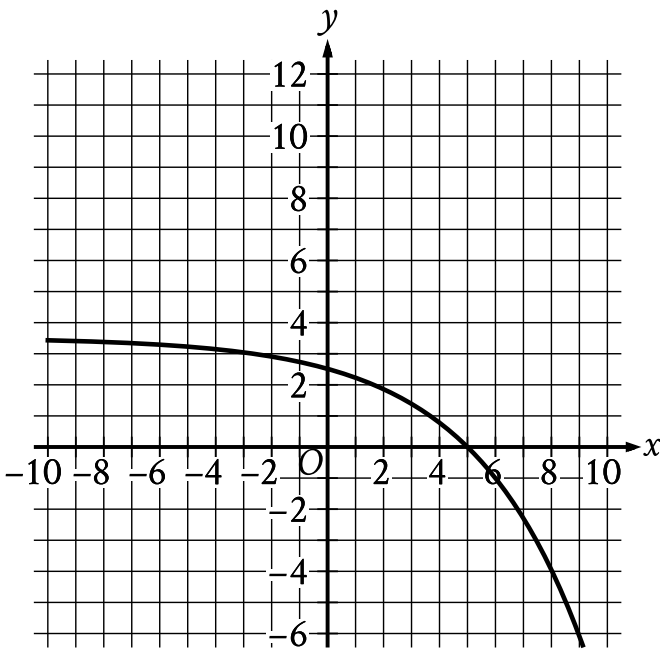
- A. $(0, 0)$
- B. $(0, 4)$
- C. $(4, 5)$
- D. $(5, 0)$

Question ID 79e6ec70

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 79e6ec70

SAT5 M 2.3



What is the x-intercept of the graph shown?

- A. $(-5, 0)$
- B. $(5, 0)$
- C. $(-2, 0)$
- D. $(2, 0)$

Question ID c943acba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: c943acba

SAT5 M 2.4

On a street with **7** houses, **2** houses are blue. If a house from this street is selected at random, what is the probability of selecting a house that is blue?

- A. $\frac{1}{7}$
- B. $\frac{2}{7}$
- C. $\frac{5}{7}$
- D. $\frac{7}{7}$

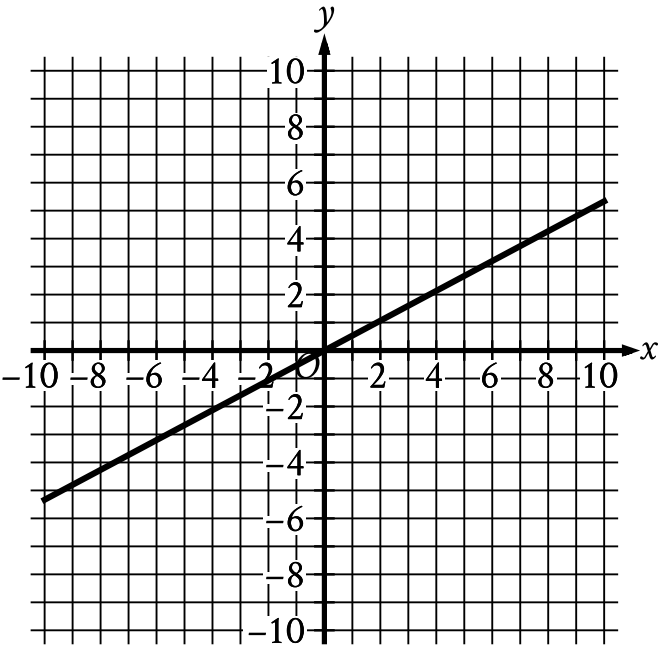
Question ID c141366d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: c141366d

SAT5 M 2.5

The graph of function f is shown, where $y = f(x)$.



Which of the following describes function f ?

- A. Increasing linear
- B. Decreasing linear
- C. Increasing exponential
- D. Decreasing exponential

Question ID 83fb222d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 83fb222d

SAT5 M 2.6

If $6n = 12$, what is the value of $n + 4$?

Question ID 361f97c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 361f97c7

SAT5 M 2.7

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

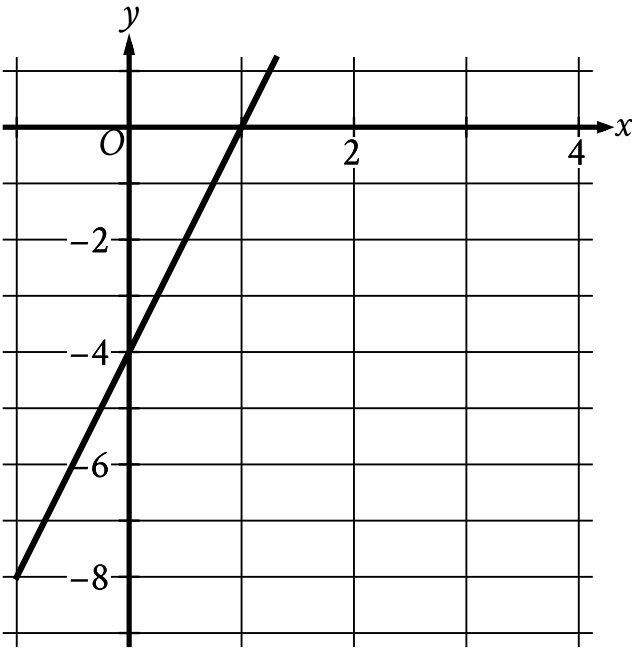
- A. ~~−30~~
- B. ~~37~~
- C. ~~40~~
- D. ~~43~~

Question ID 33e4af6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 33e4af6b

SAT5 M 2.8



The graph of the function f is shown, where $y = f(x)$. What is the y -intercept of the graph?

- A. $(0, -1)$
- B. $(0, -4)$
- C. $(0, 1)$
- D. $(0, 4)$

Question ID cdec4c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: cdec4c87

SAT5 M 2.9

$y = 12x - 20$ $y = 28$ What is the solution (x, y) to the given system of equations?

- A. $(4, 28)$
- B. $(20, 28)$
- C. $(28, 4)$
- D. $(28, 20)$

Question ID f52123e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: f52123e0

SAT5 M 2.10

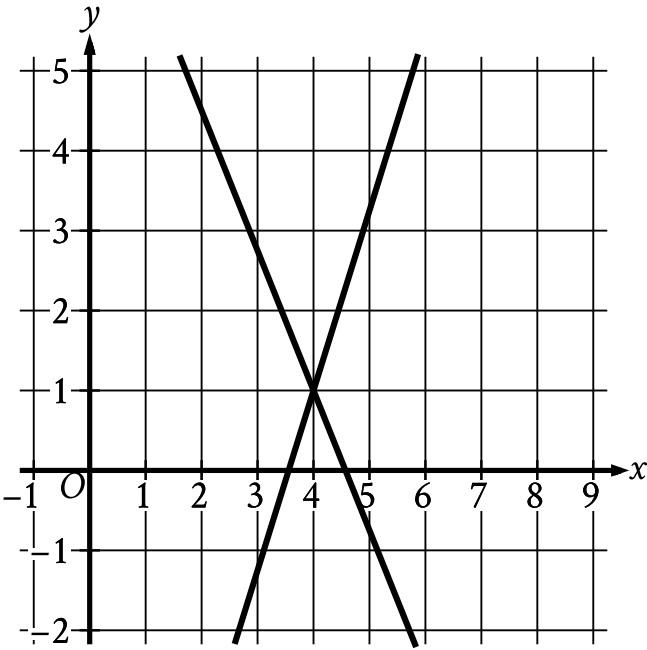
23, 27, 27, 32, 35, 36, 52 What is the range of the 7 scores shown?

Question ID 4b06557b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 4b06557b

SAT5 M 2.11



The graph of a system of linear equations is shown. The solution to the system is (x, y) . What is the value of x ?

Question ID 94ff3e2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 94ff3e2d

SAT5 M 2.12

The function h is defined by $h(x) = \frac{8}{5x+6}$. What is the value of $h(2)$?

Question ID 979b0b8d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 979b0b8d

SAT5 M 2.13

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 39 and passes through the point $(0, 0)$. Which equation defines f ?

- A. $f(x) = -39x$
- B. $f(x) = \frac{1}{39}x$
- C. $f(x) = x - 39$
- D. $f(x) = 39x$

Question ID 9e5863bd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 9e5863bd

SAT5 M 2.14

For a snowstorm in a certain town, the minimum rate of snowfall recorded was **0.6** inches per hour, and the maximum rate of snowfall recorded was **1.8** inches per hour. Which inequality is true for all values of *s*, where *s* represents a rate of snowfall, in inches per hour, recorded for this snowstorm?

- A. $s \geq 2.4$
- B. $s \geq 1.8$
- C. $0 \leq s \leq 0.6$
- D. $0.6 \leq s \leq 1.8$

Question ID bc638f2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bc638f2d

SAT5 M 2.15

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

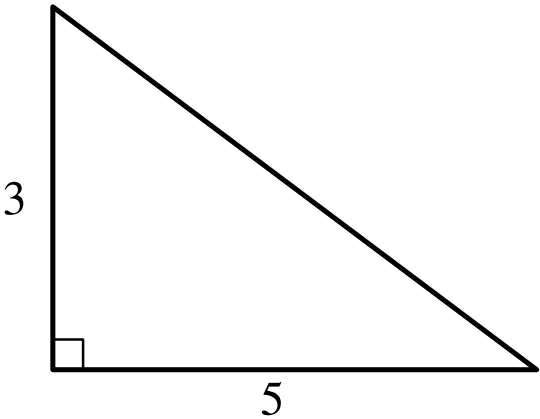
- A. Tavon will keep the vine plant for 9 months.
- B. The vine plant is expected to grow 9 inches each month.
- C. The vine plant is expected to grow to a maximum length of 9 inches.
- D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

Question ID a4ed5285

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a4ed5285

SAT5 M 2.16



Note: Figure not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

Question ID bbb0359a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: bbb0359a

SAT5 M 2.17

x	1	2	3
y	11	16	21

The table shows three values of x and their corresponding values of y . Which equation represents the linear relationship between x and y ?

- A. $y = 5x + 6$
- B. $y = 5x + 11$
- C. $y = 6x + 5$
- D. $y = 6x + 11$

Question ID 30a07668

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 30a07668

SAT5 M 2.18

$y = 4x$ $y = x^2 - 12$ A solution to the given system of equations is (x, y) , where $x > 0$. What is the value of x ?

Question ID c3d78831

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c3d78831

SAT5 M 2.19

At a particular track meet, the ratio of coaches to athletes is **1** to **26**. If there are x coaches at the track meet, which of the following expressions represents the number of athletes at the track meet?

- A. $\frac{x}{26}$
- B. $26x$
- C. $x + 26$
- D. $\frac{26}{x}$

Question ID 161126cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 161126cf

SAT5 M 2.20

$f(x) = (1.84)^{\frac{x}{4}}$

The function f is defined by the given equation. The equation can be rewritten as $f(x) = \left(1 + \frac{p}{100}\right)^x$, where p is a constant. Which of the following is closest to the value of p ?

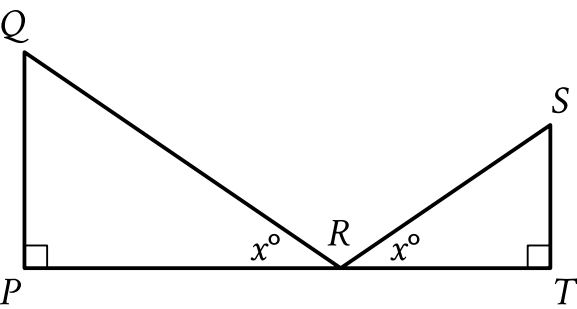
- A. 16
- B. 21
- C. 46
- D. 96

Question ID 51f26ce8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 51f26ce8

SAT5 M 2.21



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are **14**, **15**, **20**, and **25**, respectively. What is the length of $\overline{SR}$?

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

Question ID 498d6795

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 498d6795

SAT5 M 2.22

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$